

SYSTEM ANALYSIS & DESIGN

Network Vulnerability Assessment and Hardening

1. System Analysis

This phase provides a comprehensive breakdown of the project scope, identifying the necessary requirements to transition from a conceptual security objective to a functional, secure technical environment.

1.1 Project Objective

The primary goal is to establish a simulated network environment to conduct a full lifecycle of security operations: from initial reconnaissance and vulnerability identification to the implementation of defensive hardening and final validation via penetration testing.

1.2 Requirements Analysis

Functional Requirements:

- **Network Discovery:** Identification of all live assets, hostnames, and IP configurations within the target subnet.
- **Scanning Capabilities:** Deep inspection of services, protocols, and versions to map against known CVE databases.
- **Mitigation Framework:** Technical ability to apply system-level and network-level security controls.
- **Security Validation:** A mechanism to test the efficacy of applied hardening via ethical exploitation.

Non-Functional Requirements:

- **Isolation:** The project must be confined to a virtual lab to prevent any external network interference.

- **Reversibility:** Implementation of snapshots to allow system recovery in case of misconfiguration during hardening.
- **Documentation:** Detailed logging of "Before" and "After" security states for audit purposes.

2. System Design

2.1 Procedural Design (4-Week Project Workflow)

The system is designed following a structured, sequential methodology where each phase serves as a foundation for the subsequent week:

- **Week 1: Reconnaissance Phase**

Design of scanning scripts and discovery methods to map the attack surface. This includes passive info gathering and active host discovery.

- **Week 2: Vulnerability Analysis Phase**

Logical configuration of vulnerability scanners (Nessus/OpenVAS). Design of scanning policies to detect misconfigurations, outdated software, and weak credentials.

- **Week 3: Hardening & Mitigation Phase**

Design and application of technical security controls. This includes disabling non-essential services, configuring host-based firewalls, and applying OS-level security benchmarks.

- **Week 4: Validation & Reporting Phase**

Execution of a controlled penetration test to verify that the hardening measures successfully mitigated the previously identified risks. Preparation of the final technical report.

3. Implementation Roadmap

Timeline	Core Task	Primary Deliverables
Week 1	Reconnaissance	Network Map & Asset Inventory.
Week 2	Vulnerability Scanning	Detailed Scan Reports & Risk Analysis.
Week 3	System Hardening	Hardening Documentation & Rescan Logs.
Week 4	Penetration Testing	Final Report & Project Presentation.

4. Constraints & Risk Management

- **Resource Allocation:** The design assumes sufficient host hardware to support multiple virtual machine instances.
- **Scope Limitation:** Hardening is prioritized for critical vulnerabilities identified in the Week 2 analysis phase.
- **Safety Protocol:** All intrusive tests are performed only on documented target machines within the isolated lab.